A stronger foundation for RL on recursive language models

Reproducing Segment Policy Optimization on GSM8K before extending RL to RLMs

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In one sentence

Rerunning the authors' SPO code on one A100 took the matched starting model from 39.8% to 58.2% on GSM8K (paper: 56.7%) at an endpoint fixed before training; the RL workflow is now a tested reference for the RLM work.

Before extending RL, establish that we can make it work.

The advisor's challenge: reproduce a published learning result before drawing conclusions from our more complex RLM experiments.

Why this matters

Reinforcement learning (RL) trains a model using feedback on its own attempts. We need evidence that our training and evaluation workflow can produce real improvement.

Our starting question

Can feedback teach a model to use tools and delegate useful subtasks?

The calibration exercise

Reproduce a known math RL result, where answers and the target score can be checked.

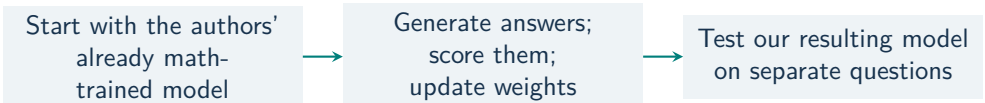
What this buys us

A tested reference workflow for separating learning problems from measurement problems.

RLM = recursive language model: a model that can use code and call models on subtasks. The reproduction tests the RL workflow, not RLM performance.

We reproduced the learning stage, not just a released model.

Experiment: Segment Policy Optimization (SPO), a published RL method.
Model and task: RhoMath 1.1B on GSM8K arithmetic word problems; one A100.



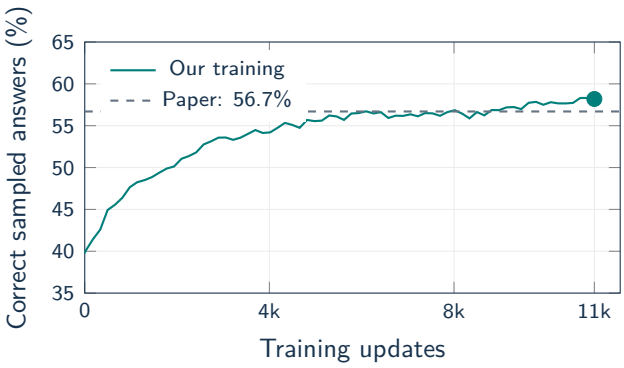
What is distinctive about SPO?

It samples possible continuations from parts of an answer to estimate which parts increase the chance of success. This gives more local feedback than a final right/wrong label alone.

Two different checks: scoring the authors' trained model checks our evaluator; training our own model tests whether we can reproduce learning.

Guo et al., Segment Policy Optimization, arXiv:2505.23564v2, Sections 3–6 and Appendix H. We reproduce RL, not pretraining or the earlier supervised learning stage.

Our own training recovered the paper's performance level.



Matched start	39.77%
Paper target	56.70%
Our final	**58.19%**
Best intermediate	58.32%

+18.42 percentage points
over our starting model.

12,281 correct / 21,104 answers.
1,319 questions, 16 answers each.

Fixed endpoint:
11,040 updates. Not selected for the best score.

Metric: mean sampled-answer correctness (mean pass@1), not at-least-one-correct. Every scheduled test is shown; final answers independently regraded. One training seed; disclosed adaptations, not exact historical replay. Sources: final audit and curve provenance in evidence.md.

The first attempt collapsed; the diagnosis is the main lesson.

What happened. Test accuracy rose to 50.5% at 4,336 updates, fell to 22.2% at 7,856, and the process crashed.

What we found. An independent regrade showed the decline was real. Generation requests began with the BOS token while training-side tokenization omitted it, so answers were sampled under one input and scored under another. The crash: a batch guard zeroed four batches in a row and the optimizer rejected the zero gradient.

What we did. Three isolated, hash-pinned fixes; a restart from the original weights; the endpoint fixed in advance.

Lessons we carry forward

- Score a known model first: it separates evaluator bugs from training bugs.
- Report the gain from a matched start under one scoring rule.
- Check that rewarded tokens are the scored tokens, BOS included.
- Regrade saved answers independently; preserve failed runs.

Answer to the advisor's challenge: we can reproduce a published LLM RL result and verify the gain. RLM is the next test.

docs/diagnosis-and-restart-20261012.md and evidence.md. The BOS mismatch is a supported hypothesis, not a demonstrated cause.

Carry the discipline into RLM; test the new assumptions.

The new challenge: an RLM must choose useful tool calls and subtasks, combine their results, and do so within a limited budget.

Question for RLM	Small next comparison *(proposed, not yet a result)*
Can the starting model use the workflow?	Direct answers versus RLM before RL, on the same tasks: check tool use, correctness and cost.
Does RL improve useful decisions?	The same RLM before versus after RL: untouched evaluation tasks, matched budgets.
Would more local feedback help?	After a basic learning signal: final-outcome rewards versus feedback on intermediate decisions.

Keep: checked rewards, matched baselines, fixed evaluation rules, saved evidence.

Do not assume: SPO or its exact settings will transfer to RLM.

We now have a credible RL reference point. Next: test learning in a recursive, tool-using environment. These comparisons are proposals, not completed RLM results.

Appendix: what exactly was reproduced and checked?

Item	Recorded protocol or limitation
Training	Authors' RhoMath 1.1B starting weights; SPO-chain interval 5; seed 42; 690 collections / 11,040 updates, original 1,000-collection learning-rate horizon.
Data and test	7,100 training / 373 validation / 1,319 test questions. 16 test samples per question; temperature 0.35; effective top-p 1.0.
Independent checks	All 33,024 final answers across three splits and 21 metrics rechecked. Final full model/optimizer checkpoint preserved with matching file hashes.
Reference checks	Authors' released model: 56.69%, versus paper 56.7%. Our starting model: 39.77%, versus paper starting reference 40.3%.
Fidelity and scope	Process-local BOS token alignment and BF16 numerical adapters disclosed. Historical equivalence unresolved. One seed; no causal ablation or measured RLM transfer.

BOS = beginning-of-sequence token; BF16 = numerical format used in training. Full protocol, immutable audit references and implementation disclosures: evidence.md and the linked final research report.